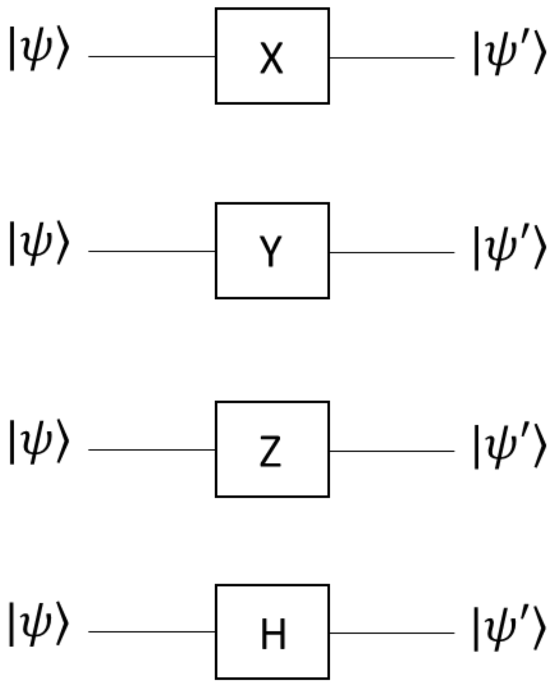
PHYS 512 (T 251)
Practice Questions

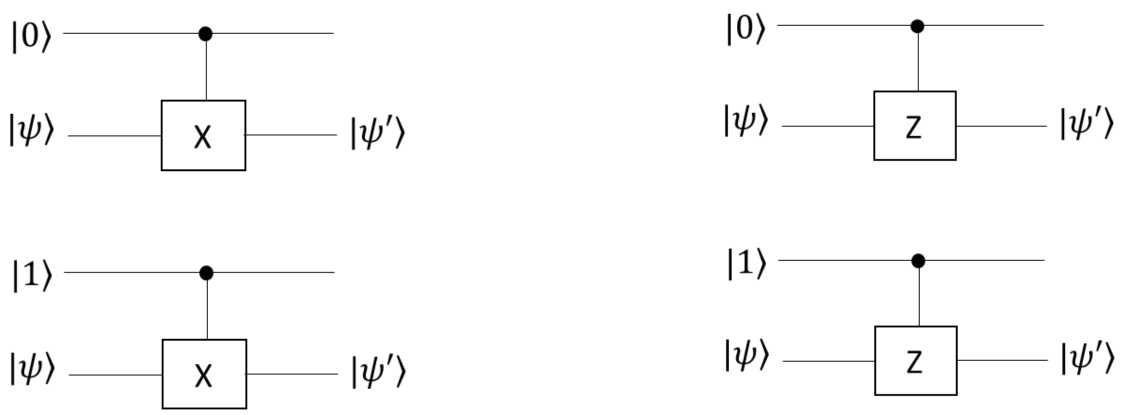
Quantum Circuits

Note: Before next lecture on Wed (30 Nov) try to solve all of the following questions related to Quantum circuits. (No need to submit)

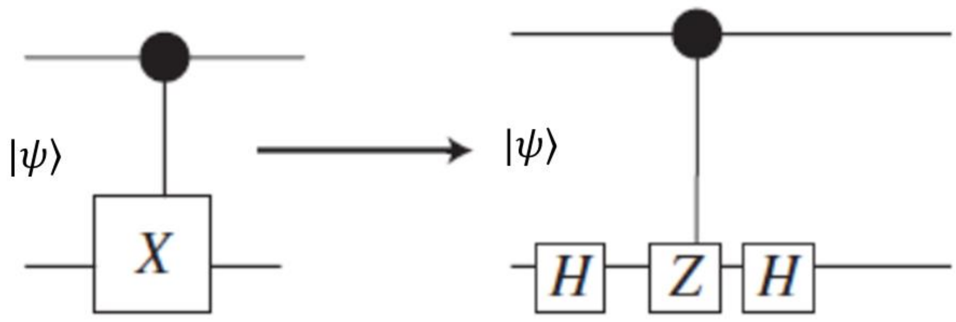
Q.1 For given input state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, for each circuit write down the output state $|\psi'\rangle$



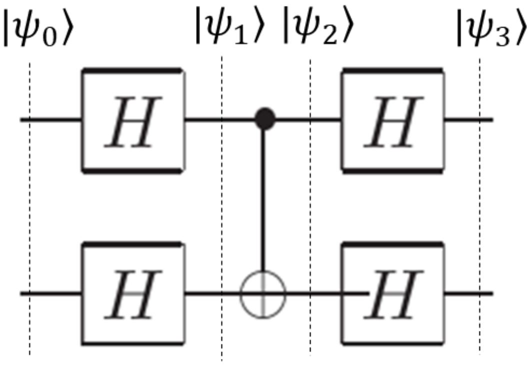
Q.2 For given input state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, for each circuit write down the output state $|\psi'\rangle$



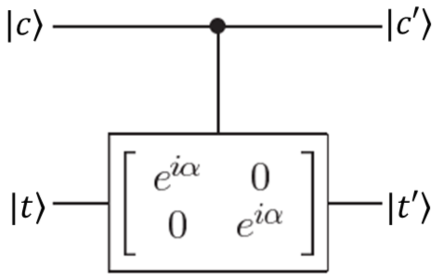
Q.3 For input states $|\psi\rangle = |00\rangle$ and $|\psi\rangle = |10\rangle$ show that output state of the both circuits (on the left and right) is same.



Q.4 For two given possible input states, $|\psi_0\rangle = |00\rangle$ and $|\psi_0\rangle = |11\rangle$, find the state $|\psi_1\rangle$, $|\psi_2\rangle$ and $|\psi_3\rangle$



Q.4 Consider the following gate, known as ***controlled phase shift gate*** (matrix form of gate is given in diagram). Complete the truth table for given circuit.



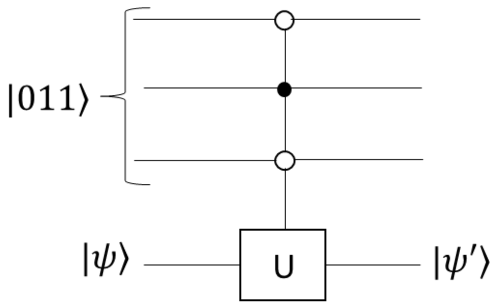
$\|c\rangle$	$\|t\rangle$	$\|c'\rangle$	$\|t'\rangle$
0	0		
0	1		
1	0		
1	1		

Q.5 For given state of control qubit $|011\rangle$, which of the following is correct for the out put

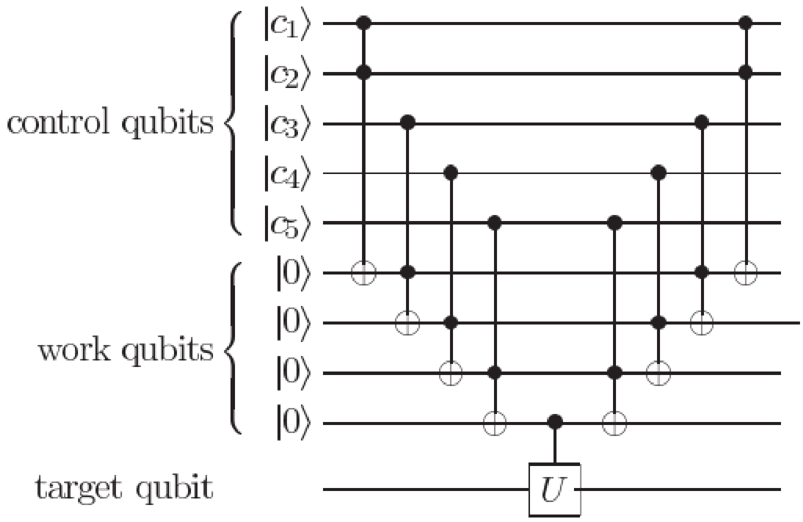
$$|\psi'\rangle = |\psi\rangle$$

Or

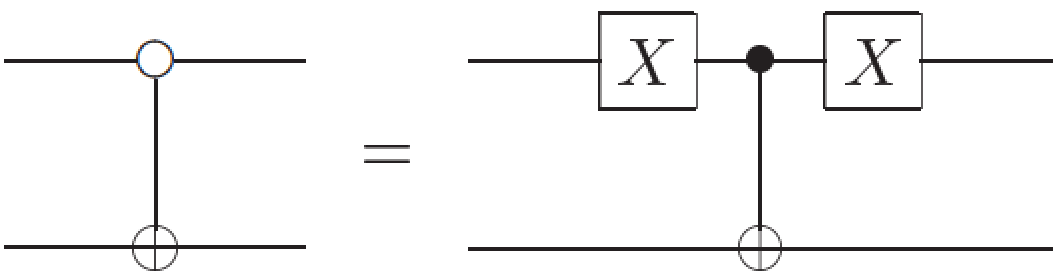
$$|\psi'\rangle = U|\psi\rangle$$



Q.6 For which combination of inputs ($|C_1\rangle$ through $|C_5\rangle$) Unitary gate U will act on target qubit (all gates are normal CNOT gate.

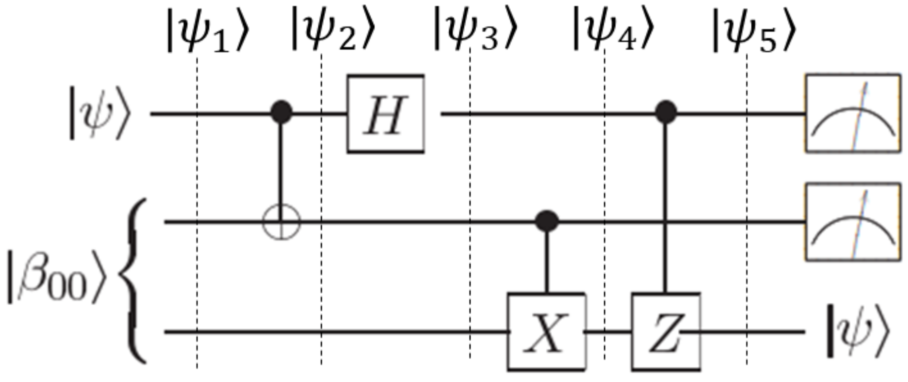


Q.7 Draw truth table for the following circuit



Q.8

In the following circuit write down all state $|\psi_1\rangle$ through $|\psi_5\rangle$. The input state $|\psi\rangle$ is general state i.e. $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$



Upon measuring the first two qubits in $|\psi_5\rangle$, third qubit will collapse (Y/N). Briefly explain your answer.

Eigen States, Time evolution and expectation Values

Q.1 Eigen states, time evolution and expectation value

A Hilbert space is spanned by an orthonormal basis$\{|1\rangle,\ |2\rangle, |3\rangle\}$. An operator

$$\hat{A} = i|1\rangle\langle 2| - i|2\rangle\langle 1| + |3\rangle\langle 3|$$

(a) Express $\hat{A}$ in matrix form
(b) Find eigenstates of $\hat{A}$ *(Hint: no need to diagonalize the full 3x3 matrix, just diagonolize the selected 2x2)*
(c) Write $\hat{A}$ in the basis of its eigenstates.
(d) The Hamiltonian of a system is given by

$$\hat{H} = \hbar\omega|1\rangle\langle 1| + 2\hbar\omega|2\rangle\langle 2| + 3\hbar\omega|3\rangle\langle 3|$$

If the state of system at $t = 0\ is\ |\psi(0)\rangle = \frac{1}{\sqrt{2}}\ |1\rangle - \frac{i}{\sqrt{3}}\ |3\rangle$,what is the sate at time t?

(Remember time evolution of state $|\psi(t)\rangle = e^{-\frac{i\hat{H}t}{\hbar}}\ |\psi(0)\rangle$)

(e) We define two observables,

$$\hat{B} = b(|1\rangle\langle 2| + |2\rangle\langle 1| + 2|3\rangle\langle 3|)$$

and

$$\hat{C} = c(2|1\rangle\langle 1| + |2\rangle\langle 3| + |3\rangle\langle 2|)$$

A measurement of $\hat{B}$ is made at time t = 0, yielding the outcomes 2b. After measurement what is the new state? Find the state after time t?

(f) Given the initial state in part (d), Find the expectation value $\langle\psi(t)|\hat{C}|\psi(t)\rangle$

Q.2 Bell States

Express all Bell states, $\beta_{00}\ and\ \beta_{10}$ in $|+\rangle, |-\rangle$

Q.3 Entangled states

$$|\psi\rangle = \left(\frac{1}{2} + \frac{i}{2}\right)\ |00\rangle + \frac{1}{2}\ |01\rangle - \frac{i}{2}\ |10\rangle$$

(a) What is the probability of measuring the state $|\psi\rangle$ in $|10\rangle$state?
(b) If $|\varphi\rangle = (I \otimes \sigma_x)|\psi\rangle$, where σ_x is Pauli matrix. Write $|\varphi\rangle$ in computational basis.
(c) Apply the operator $P_0 \otimes I$ and $I \otimes P_1$on the state $|\varphi\rangle$ (where $P_0 = |0\rangle\langle 0|$ and $P_1 = |1\rangle\langle 1|$are projection operators). Normalize the states after applying the operators.

Q.4 Entanglement States (GHZ state)

Consider the following entangled state

$$|\psi\rangle = \frac{1}{\sqrt{2}}\,|000\rangle + \frac{1}{\sqrt{2}}\,|111\rangle$$

(a) Above state is entangled?

(b) Draw the quantum circuit that can produce entangled state$|\psi\rangle$, clearly mark the state of qubits after every gate.

(c) Apply the operator $\hat{A} = \hat{X}\otimes\hat{Y}\otimes\hat{Z}$on sate $|\psi\rangle$ i.e. find the state $|\phi\rangle$

$$|\phi\rangle = \left(\hat{X}\otimes\hat{Y}\otimes\hat{Z}\right)|\psi\rangle$$

(d) What is the probability of measuring the state $|\phi\rangle$in $|+++\rangle$ state?

(e) Find the expectation value $\hat{A}$ when system was in state$|\psi\rangle$

Q.5 Measurement and State Collapse

We know state if measurement operator M_m act on state $|\psi\rangle$, the new normalized state $|\psi'\rangle$ can be expressed as

$$|\psi'\rangle \frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}}$$

(a) Consider the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}\,|00\rangle + \frac{1}{\sqrt{2}}\,|11\rangle$$

Upon measurement the state collapsed to (new state is normalized)

$$|\psi'\rangle = \ |00\rangle$$

Suggest a possible measurement operator and express it in matrix form

(b) Consider another state

$$|\psi\rangle = \frac{1}{\sqrt{3}}\,|00\rangle + \frac{1}{\sqrt{3}}\,|11\rangle + \frac{1}{\sqrt{3}}\,|01\rangle$$

Upon measurement the state collapsed to (new state is normalized)

$$|\psi'\rangle = \frac{1}{\sqrt{2}}\,|01\rangle + \frac{1}{\sqrt{2}}\,|11\rangle$$

Suggest a possible measurement operator and express it in matrix form.

(c) Consider another state

$$|\psi\rangle = \frac{\sqrt{2}+i}{\sqrt{20}}\,|000\rangle + \frac{1}{\sqrt{2}}\,|101\rangle + \frac{1}{\sqrt{10}}\,|100\rangle + \frac{i}{2}\,|010\rangle$$

Upon measurement the state collapsed to (new state is normalized)

$$|\psi'\rangle = \frac{1}{\sqrt{2}}\,|101\rangle + \frac{1}{\sqrt{2}}\,|100\rangle$$

Suggest a possible measurement operator and express it in matrix form.

Basics and Density Operator

Q.1

(i) Find the adjoint of the operator

$$\hat{A} = 2|0\rangle\langle 1| - i2|1\rangle\langle 0|$$

Is the operator $\hat{A}$ Hermitian? [2]

(ii) Find the eigenvalues for the following matrix [2]

$$\hat{T} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

(iii) Find the expectation value of the observable Z_1Y_2(where Z_1 and Y_2 are Pauli operators of qubit 1 and 2, respectively) for a two-qubit system measured in state $|\psi\rangle$, where [3]

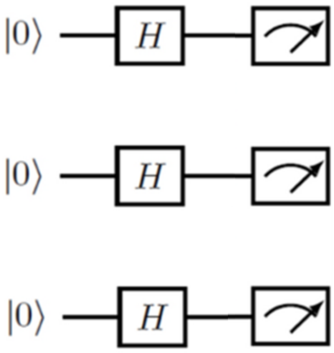
$$|\psi\rangle = \frac{1}{\sqrt{2}}(\,|01\rangle + |10\rangle)$$

(iv) Show that the operator

$$U = cos\theta|0\rangle\langle 0| + e^{i\phi}sin\theta|0\rangle\langle 1| + e^{-i\phi}sin\theta|1\rangle\langle 0| - cos\theta|1\rangle\langle 1|$$

is unitary. Where θ and ϕ are real parameters. [3]

(v) For the following circuit find the probability that the measurement apparatus detects the state $|010\rangle$. [2]



(vi) Suppose

$$\rho = \begin{pmatrix} \frac{1}{3} & \frac{-i}{3} \\ \frac{i}{3} & \frac{2}{3} \end{pmatrix}$$

Diagonal elements (i.e. $\frac{1}{3}$ and $\frac{2}{3}$) are eigenvalues of ρ (**Yes/No**) [1]

(vii) Consider *an ensemble* in which 40% of the systems are known to be prepared in state

$$|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$$

and 60% of the systems are prepared in the state

$$|\phi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

a) Determine the density operator for *the ensemble* in computational basis. [3]

b) A member of the ensemble is drawn, what are the probabilities of finding it in the state $|0\rangle$ [2]

(viii) Consider the following Bell state

$$|\beta_{10}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

where the first qubit is of Alice and second qubit is of Bob

a) Find the density matrix $\rho_{AB} = |\beta_{10}\rangle\langle\beta_{10}|$ [2]

b) Find the find the density matric ρ_A, of Alice qubit [2]

$$\rho_A = Tr_B(\rho_{AB})$$

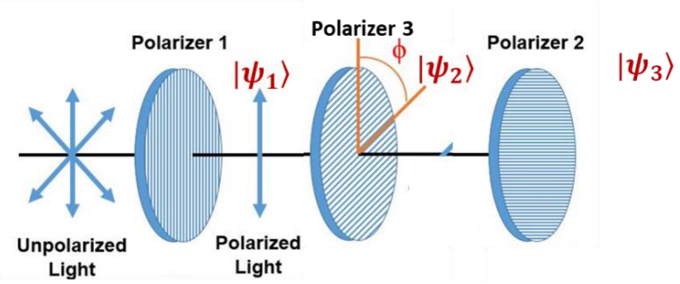
c) Show that $|\beta_{10}\rangle$is an entangled state [2]

(ix) Which quantum gate(s)/operator can convert the state $\frac{2}{3}|00\rangle + \frac{\sqrt{5}}{3}|11\rangle$ to $\frac{2}{3}|10\rangle - \frac{\sqrt{5}}{3}|01\rangle$ [2]
(just give name of the gate(s)/operator in proper notation)

(x) In Fig given below, input quantum state of un-polarized light is $|\psi\rangle$. Write down the ***normalized quantum states*** $|\psi_1\rangle, |\psi_2\rangle, and\ |\psi_3\rangle$, when ϕ=45º. [3]
(Assume vertically polarized light is $|1\rangle$)

Input state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$



(xi) Write down the name of three possible physical qubits. [1]

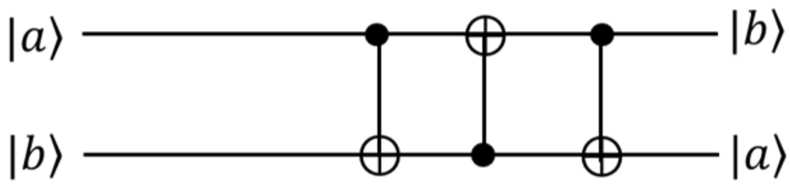
(xii) Verify that the Bell basis (states) forms an orthonormal basis for the two-qubit state space. (Just verify for ***any pairs*** of Bell states) [4]

(xiii) For given

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Find $(H\otimes I)CNOT|\psi\rangle = ?$ [3]

(xiv) Show that the following circuit swaps the input. [3]



(xv) Draw a circuit that can create a Bell states. Draw the circuit to create only one Bell state (of your choice) with input $|00\rangle$. [2]

(xvi) Show that $\sigma_x \otimes \sigma_x$ commutes with $\sigma_z \otimes \sigma_z$ [4]

(xvii) Consider the following two quantum states in computational basis [4]

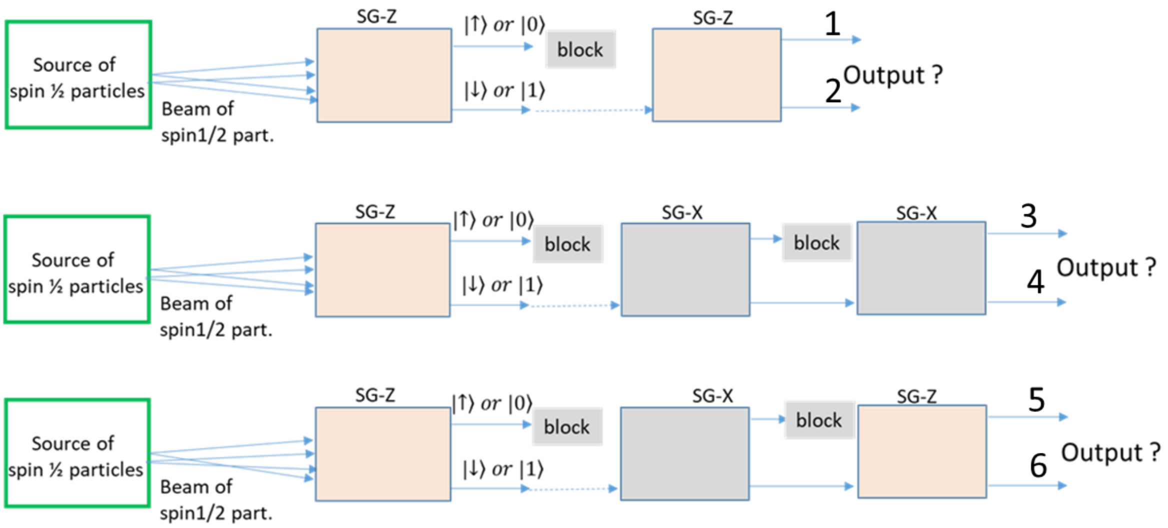
$$|\psi\rangle = \frac{1}{\sqrt{2}}\Big|0\rangle - \frac{i}{\sqrt{2}}\Big|1\rangle, \qquad |\varphi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$$

Compute the following

(a) $|\psi\rangle \otimes |\varphi\rangle$

(b) $(\sigma_x \otimes \sigma_z)(|\psi\rangle \otimes |\varphi\rangle)$

(xviii) Source of spin-1/2 particle produces only ONE spin-1/2 particle at a time. You repeat this 'single spin-1/2 particle experiment' 1000 times, how many times you will detect a spin-1/2 particle at output stages 1, 2, 3, 4, 5, and 6 (just write down the number) [3]

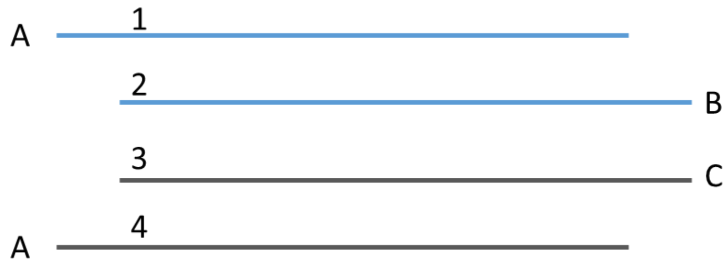


Q.2 [10 Points (4+6)] Entanglement swapping

A (Alice) share two different entangled states with B (Bob) and C (Charlie) as shown in the figure. Qubit 1 and 4 belong to Alice and 2 to Bob and 3 to Charlie. (Blue lines represent 1 and 2 entangled states and black lines represents 3 and 4 entangled states).

$$|\beta_{00}\rangle_{12} = \frac{1}{\sqrt{2}}(|00\rangle_{12} + |11\rangle_{12})$$

$$|\beta_{00}\rangle_{34} = \frac{1}{\sqrt{2}}(|00\rangle_{34} + |11\rangle_{34})$$



Combined quantum state of above two entangled qubits, 1,2 and 3,4 can be expressed as

$$|\beta_{00}\rangle_{12} \otimes |\beta_{00}\rangle_{34} = \left(\frac{1}{\sqrt{2}}(|00\rangle_{12} + |11\rangle_{12})\right) \otimes \left(\frac{1}{\sqrt{2}}(|00\rangle_{34} + |11\rangle_{34})\right)$$

Above expression simplified in way where Alice qubit separated (factorized) from Bob and Charlie qubits is

$$|\beta_{00}\rangle_{12} \otimes |\beta_{00}\rangle_{34} = \frac{1}{2}[|\beta_{00}\rangle_{14} \otimes |\beta_{00}\rangle_{23} + |\beta_{01}\rangle_{14} \otimes |\beta_{01}\rangle_{23} + |\beta_{10}\rangle_{14} \otimes |\beta_{10}\rangle_{23} + |\beta_{11}\rangle_{14} \otimes |\beta_{11}\rangle_{23}]$$

Where $|\beta_{00}\rangle, |\beta_{01}\rangle, |\beta_{10}\rangle$ *and* $|\beta_{11}\rangle$ are Bell states (for details check the formula-sheet)

a) If Alice measures $|\beta_{10}\rangle$,

(i) write down the Bob and Charlie combined state in computational basis.

(ii) If Bob measures $|1\rangle$, what Charlie will measure?

b) If Alice measures $|\beta_{11}\rangle$,

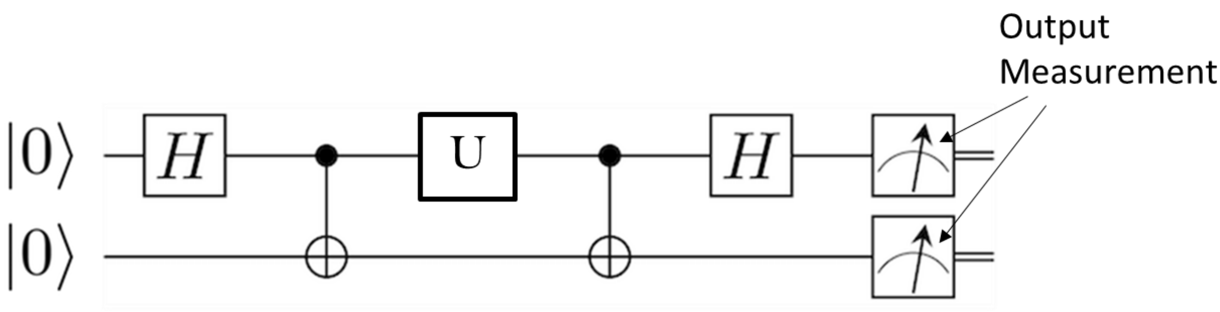
(i) write down the Bob and Charlie combined state in computational basis.

(ii) What is the probability of measuring $|1\rangle$ by the Bob?

(iii) If Bob measures $|0\rangle$, what is the new normalized state of ABC's quantum circuit?

Q.3 [12 points (4+4+4)**] Super-Dense Coding**

Following is the super-dense coding quantum circuit.



a) Write down the equation of above quantum circuit (equation should include H, U, CNOT or any other required operation/gate(s)). [4]

b) If unitary gate-U is a Pauli X-gate, find the meters reading (output) of the quantum circuit (Show output at each stage of the circuit). [4]

c) If unitary gate-U is a Pauli ZX-gate, find the meters reading (output) of the quantum circuit (Show output after U/ZX-gate of the circuit). [4]

Q.4 [12 Points (8+4)] ***Quantum teleportation***

Alice and Bob each possess one member of a pair of interacting magnetic diploes (spin ½ particles). The interaction Hamiltonian for two magnetic dipoles separated by a distance r is given by

$$H_I = \frac{\mu^2}{r^3}(\vec{\sigma}_A . \vec{\sigma}_B - 3Z_A \otimes Z_B)$$

Where $\vec{\sigma}_A$ and $\vec{\sigma}_B$ are Pauli operators for Alice and Bob, respectively.
[Note: $\vec{\sigma}_A . \vec{\sigma}_B = X_A \otimes X_B + Y_A \otimes Y_B + Z_A \otimes Z_B$]

(g) Show that the matrix form of the interaction Hamiltonian (H_I) is

$$H_I = \frac{\mu^2}{r^3}\begin{pmatrix} -2 & 0 & 0 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

(h) Verify that the following Bell stats are eigenstates of H_I

$$|\beta_{01}\rangle = \frac{1}{\sqrt{2}}(\,|01\rangle + \,|10\rangle)$$
$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(\,|00\rangle + \,|11\rangle)$$

Q.7 [10 Points (5+5**)]** ***Trace distance and Fidelity***

Consider the density matrices $\rho, \sigma, and\ \chi$

$$\rho = \frac{3}{4}|+\rangle\langle+| + \frac{1}{4}|-\rangle\langle-|$$

$$\sigma = |\psi\rangle\langle\psi|, \text{ where } |\psi\rangle = \frac{i}{\sqrt{5}}|0\rangle + \frac{2}{\sqrt{5}}|1\rangle$$

$$\chi = |\varphi\rangle\langle\varphi|, \text{ where } |\varphi\rangle = \left(\frac{1}{2} + \frac{i}{2}\right)|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

a) Express ρ, σ, and χ in matrix form (using $\{|0\rangle, |1\rangle\}$ basis)

b) Compute

$$\delta(\rho,\sigma) = \frac{1}{2}Tr|\rho - \sigma|$$

(Note: $\delta(\rho,\sigma)$ is called trace distance)

Additional Information

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$; $\quad Y = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

$i\hbar\frac{\partial}{\partial t}|\psi\rangle = H|\psi\rangle; \qquad |\psi(t)\rangle = e^{-iHt/\hbar}|\psi(0)\rangle$

$\hat{n} = (sin\theta cos\varphi, sin\theta sin\varphi, cos\theta);$

$\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle; \qquad S.D = \sqrt{\langle\sigma^2\rangle - \langle\sigma\rangle^2}$

$|\psi\rangle = \begin{pmatrix} cos\theta/2 \\ e^{i\varphi}sin\theta/2 \end{pmatrix}$

$e^{i\theta Z} = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}$

Bell States:

$|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$; $|\beta_{01}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$; $|\beta_{10}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$; $|\beta_{11}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

Bloch Vector $\vec{S}$ for density matrix-ρ: $|\vec{S}| = \sqrt{s_x^2+s_y^2+s_z^2}$, where $s_x = Tr(X\rho)$; $s_y = Tr(Y\rho)$; $s_z = Tr(Z\rho)$

Trace Distance: $\delta(\rho,\sigma) = \frac{1}{2}Tr|\rho - \sigma|$; $\delta(\rho,\sigma) = \sqrt{1 - \langle\psi|\sigma|\psi\rangle}$ (if ρ is a pure state);

$\delta(\rho,\sigma) = \frac{1}{2}Tr|\sum_i(r_i - s_i)\,|u_i\rangle\langle u_i|$;

$\delta(\rho,\sigma) = \frac{1}{2}|\vec{r} - \vec{s}|, where\ \vec{r}\ and\ \vec{s}$ are Bloch vector corresponding to $\rho\ and\ \sigma$, respectively.

Fidelity: $F(\rho,\sigma) = Tr\left(\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\right) = \sum_i\sqrt{r_i s_i}$ where $r_i\ and\ s_i$are eigenvalues of $\rho\ and\ \sigma$, respectively.

CNOT operation: $|a\ b\rangle \rightarrow |a\ b\oplus a\rangle$

$i\hbar\frac{\partial}{\partial t}|\psi\rangle = H|\psi\rangle; \qquad |\psi(t)\rangle = e^{-iHt/\hbar}|\psi(0)\rangle$

$\hat{n} = (sin\theta cos\varphi, sin\theta sin\varphi, cos\theta);$

$\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle; \qquad S.D = \sqrt{\langle\sigma^2\rangle - \langle\sigma\rangle^2}$

$|\psi\rangle = \begin{pmatrix} cos\theta/2 \\ e^{i\varphi}sin\theta/2 \end{pmatrix}$

$$e^{i\theta Z} = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}$$

***Useful expression related to Density Operator* ρ**

$$\rho = |\psi\rangle\langle\psi|$$
$$\rho = \sum_{i=1}^{n} p_i \, |\psi_i\rangle\langle\psi_i|$$
$$\langle A\rangle = \langle\psi|A|\psi\rangle = Tr(\rho A)$$

Density operator *in matric form*

$$\rho = \begin{pmatrix} \langle u_1|\rho|u_1\rangle & & \langle u_1|\rho|u_n\rangle \\ & & \\ \langle u_n|\rho|u_1\rangle & & \langle u_n|\rho|u_n\rangle \end{pmatrix}$$

Time evolution *of* $\boldsymbol{\rho}$

$$\rho = \frac{-i}{\hbar}[H, \rho]$$
$$\rho(t) = U\rho(t_0)U^{\dagger}$$

***Measurement Operator* M**

$$\boldsymbol{M} = \sum_i m_i \, |m_i\rangle\langle m_i|$$

Where $|m_i\rangle$is an eigenstate with eigenvalue m_i , obeys the closure relation

$$\sum_i |m_i\rangle\langle m_i| = \mathbb{1}$$

Probability *of measuring a state* $|m_i\rangle$

$$p_{(|m_i\rangle)} = Tr(\rho M_i)$$

Where $M_i = |m_i\rangle\langle m_i|$

Probability of measuring a state $|0\rangle$ (with projection operator $P_0 = |0\rangle\langle 0|$

$$p_{(|0\rangle)} = Tr(\rho P_0) = Tr(\rho|0\rangle\langle 0|) = \langle 0|\rho|0\rangle$$

State after measurement*: In a closed system* if $M_m$represent a measuring device (measurement operator) whose eigenvalue is m, and if a measurement on the system yields m, then after measurements

$$\rho' = \frac{M_m \rho M_m^{\dagger}}{Tr\left(M_m^{\dagger} M_m \rho\right)}$$

Above expression in terms of projection operator P_n

$$\rho' = \frac{P_n \rho P_n}{Tr(P_n \rho)}$$

Competently mixed States

In a completely mixed state, the probability for the system to in each state is identical, in scuh cases

$$\rho = \frac{1}{n}\mathbb{1}$$

Since $\mathbb{1}^2 = \mathbb{1}$, we have $\rho^2 = \frac{1}{n^2}\mathbb{1}$

For n-dimension Tr($\mathbb{1}$) $= n$, therefore for competently mixed state

$$Tr(\rho^2) = Tr\left(\frac{1}{n^2}\mathbb{1}\right) = \frac{1}{n}$$

Mixed and pure state using concept of Bloch Vector

$$\rho = \frac{1}{2}\left(\mathbb{1} + \vec{S}.\vec{\sigma}\right)$$

Where $\vec{S}$ is called Bloch Vector

$\left|\vec{S}\right| = 1$for pure state

$\left|\vec{S}\right| < 1$for mixed state

$$\vec{S} = S_x\hat{x} + S_y\hat{y} + S_z\hat{z}$$

and

$$S_x = Tr(\rho X), \quad S_y = Tr(\rho Y), \quad S_z = Tr(\rho Z)$$